

Varsha V

Senior Data Engineer



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| Summary

Senior Data Engineer with 7+ years of experience designing and building scalable cloud data platforms, ETL/ELT pipelines, and enterprise data solutions. Strong expertise in Azure Databricks, Azure Data Factory, PySpark, Python, SQL, Azure Synapse Analytics, ADLS Gen2, and Delta Lake. Experienced in batch and streaming data pipelines, data warehousing, dimensional modeling, data quality, reconciliation, and performance optimization. Skilled in CI/CD, Azure DevOps, Terraform, and secure cloud data solutions supporting analytics, reporting, and business intelligence.

| Technical Skills

Azure Cloud: Azure Data Factory (ADF), Azure Databricks, Azure Synapse Analytics, Azure Data Lake Storage Gen2 (ADLS Gen2), Azure SQL Database, Azure Monitor, Azure Log Analytics, Azure DevOps, Managed Identities

GCP Cloud: BigQuery, GCP Dataflow, Pub/Sub, Cloud Storage, Cloud Build.

Programming & Query: Java, Python, PySpark, SQL, KQL (Kusto Query Language), REST APIs.

Data Engineering: ETL/ELT Pipelines, Batch Processing, Streaming Data Pipelines, Data Ingestion, Data Transformation, Distributed Data Processing

Data Architecture: Snowflake, Data Modeling, Dimensional Modeling, Medallion Architecture, Delta Lake, Data Validation & Reconciliation

Orchestration & CI/CD: Apache Airflow, Azure Data Factory Pipeline, Terraform, ARM Templates, Git, CI/CD Pipelines

Reporting & BI: Power BI, Operational Reporting, KPI Reporting, Business Intelligence, Compliance Analytics

Security & Governance: RBAC, IAM Policies, Service Principals

Operations & Process: Pipeline Monitoring, Alerting & SLA Tracking, Production Support, Performance Tuning, Root Cause Analysis, Technical Documentation, Agile/Scrum, Relational Databases, Testing.

AI and Generative: Azure OpenAI Service, Langchain, RAG, Azure AI Search, LLM, Log Analytics.

| Education

Bachelors in Computer Science and Engineering

Sep 2015-May 2019

Jawaharlal Nehru Technological University

| Certifications

Microsoft Certified: Azure Administrator Associate

| Work experience

Senior Data Engineer

Aug 2024 - present

USAA, San Antonio, TX

Working on enterprise-scale Azure cloud data engineering solutions supporting insurance operations, banking analytics, customer reporting, risk analysis, and operational intelligence platforms. Responsible for designing and developing scalable batch and near real-time data pipelines using Azure-native services to process customer transactions, policy data, claims data, and business-critical reporting datasets. Built centralized data lake and analytics platforms leveraging Medallion Architecture, Delta Lake, and Azure Synapse to support regulatory reporting, operational dashboards, and advanced analytics initiatives across business domains.

Responsibilities

- Designed and implemented scalable Medallion Architecture using Bronze, Silver, and Gold layers with Delta Lake and ADLS Gen2 to support enterprise data processing, governed data storage, and analytics-ready data warehouse solutions.
- Built enterprise data ingestion frameworks using Azure Data Factory (ADF) to load structured and semi-structured data from REST APIs, Azure SQL Database, flat files, external vendor systems, and cloud storage platforms.
- Designed and developed scalable ETL/ELT data pipelines using Azure Databricks, PySpark, Python, SQL, and Azure Data Factory to process and integrate data from multiple enterprise sources.
- Developed near real-time and streaming data pipelines using Azure Event Hubs, Azure Stream Analytics, and Databricks Structured Streaming for customer activity tracking and operational event processing.
- Designed and implemented curated data marts, dimensional data models, and data warehouse structures in Azure Synapse Analytics to support efficient data storage and retrieval, compliance reporting, KPI dashboards, customer analytics, and operational intelligence initiatives.
- Optimized Apache Spark workloads, data processing flows, and storage performance using partitioning, caching, broadcast joins, adaptive query execution, and performance tuning techniques to improve processing efficiency, scalability, and reduce execution time.
- Implemented automated data quality, validation, and reconciliation frameworks using PySpark and SQL for source-to-target validation, schema checks, null handling, duplicate detection, data consistency, and reliable data processing.
- Developed reusable metadata-driven Azure Data Factory pipelines to automate data ingestion, transformation, orchestration, scheduling, and deployment across development and production environments.
- Designed and maintained scalable data integration solutions using Azure Databricks, Python, and REST APIs to ingest, transform, and deliver enterprise data for analytics, reporting, and downstream business applications.
- Built CI/CD pipelines using Azure DevOps, Terraform, ARM Templates, and Git for automated code deployment and infrastructure provisioning.
- Configured RBAC, Managed Identities, Azure Key Vault integration, and Service Principals to implement data security measures, access controls, data protection protocols, and enterprise compliance standards across Azure resources.
- Collaborated cross-functionally with data scientists, data engineers, data architects, analysts, business stakeholders, and product teams to understand data requirements and deliver scalable data integration, processing, analytics, and data-driven solutions.
- Developed automated monitoring, alerting, and operational support solutions for production data pipelines using Azure Monitor, Log Analytics, and Databricks to improve reliability, SLA tracking, and issue resolution.
- Mentored junior data engineers on PySpark, SQL, Azure Databricks, Azure Data Factory, data pipeline development, troubleshooting, coding standards, and cloud data engineering best practices while supporting production deployments and issue resolution.

Environment: Azure Data Factory (ADF), Azure Databricks, Azure Synapse Analytics, Azure Data Lake Storage Gen2 (ADLS Gen2), Delta Lake, Azure Event Hubs, Azure Stream Analytics, Azure SQL Database, Azure Monitor, Log Analytics, Application Insights, PySpark, Python, SQL, Databricks Workflows, Azure DevOps, Terraform, ARM Templates, Git, CI/CD Pipelines, ETL/ELT, Data Modeling, Dimensional Modeling, Data Warehousing, REST APIs, Managed Identities, Azure Key Vault, RBAC, Medallion Architecture, Agile/Scrum

Enterprise Data Engineer

Jul 2023 - Aug 2024

Dynatrace, Detroit, MI

Worked on cloud-based data engineering and analytics solutions supporting telemetry monitoring, operational reporting, compliance analytics, and application observability platforms. Built batch and near real-time data pipelines using Azure services to process monitoring logs, telemetry data, and operational datasets from enterprise systems.

Responsibilities

- Designed, developed, and maintained scalable ETL/ELT data pipelines using PySpark, SQL, Apache Spark, and Azure Databricks to support efficient data integration, processing, and analytics of telemetry, monitoring, and operational datasets.
- Designed and implemented scalable data architecture using Medallion Architecture (Bronze, Silver, and Gold), Delta Lake, and ADLS Gen2 to support structured data processing, efficient data storage, and analytics solutions.
- Developed ingestion workflows using Azure Data Factory to load data from REST APIs, SQL databases, monitoring platforms, and cloud storage systems.
- Processed and transformed telemetry data in Databricks to support operational dashboards, reporting, and monitoring analytics.
- Optimized Spark jobs using partitioning, caching, and query tuning techniques to improve processing performance.
- Integrated Microsoft Intune and Azure Monitor data into Log Analytics for compliance reporting and anomaly detection using KQL queries.
- Developed curated datasets, dimensional data models, and data warehouse structures to support efficient data storage, retrieval, KPI reporting, and operational analytics. Implemented data quality and consistency checks using PySpark and SQL for schema validation, duplicate handling, reconciliation, and reliable data processing.
- Built CI/CD pipelines using Azure DevOps, Git, Terraform, and ARM Templates for deployment automation.
- Configured RBAC, Managed Identities, and Service Principals for secure access across Azure resources.

Collaborated cross-functionally with data engineers, analytics teams, platform engineers, data stakeholders, and business users to understand data requirements and deliver scalable data integration and analytics solutions. Developed monitoring and alerting solutions for data pipeline failures, performance issues, SLA tracking, and

production support using Azure Monitor and Log Analytics

Environment: Azure Data Factory (ADF), Azure Databricks, Azure Synapse Analytics, ADLS Gen2, Delta Lake, Azure Monitor, Log Analytics, KQL, Azure SQL Database, PySpark, Python, SQL, Azure DevOps, Terraform, ARM Templates, Git, CI/CD, ETL/ELT, Data Modeling, REST APIs, RBAC, Service Principals

Data Engineer

Mar 2022 - Jun 2023

Cummins Inc, Columbus, IN

Worked on enterprise-scale GCP cloud data engineering solutions supporting manufacturing operations, IoT equipment monitoring, supply chain analytics, and operational reporting platforms. Designed and developed scalable batch and near real-time data pipelines using Google Cloud Platform services to process IoT telemetry, ERP transactions, machine logs, and production datasets for analytics, KPI reporting, and operational intelligence initiatives.

Responsibilities

- Developed and maintained scalable ETL/ELT data pipelines using Java, PySpark, Python, SQL, Apache Spark, and Google Cloud Dataflow for efficient data integration and processing of manufacturing, operational, and supply chain datasets.
- Built batch and streaming data pipelines using GCP Dataflow, Pub/Sub, and Dataproc to process IoT telemetry, machine logs, and production event data.
- Designed and implemented cloud-native data lake and data warehouse architecture using Google Cloud Storage (GCS) and BigQuery for centralized data storage, efficient retrieval, analytics, and enterprise reporting.
- Developed Java-based ingestion and ETL components integrating ERP systems, relational databases, REST APIs, flat files, and cloud storage into BigQuery and GCS for manufacturing and enterprise analytics.
- Developed and optimized Java-based Spark applications using Dataproc to process and transform large-scale manufacturing and inventory datasets.
- Designed and developed curated datasets, reporting tables, dimensional data models, fact and dimension tables, and scalable data warehouse solutions in BigQuery for dashboards, KPI reporting, and business analytics.
- Implemented data quality, validation, consistency, and reconciliation frameworks using PySpark and SQL for schema validation, duplicate handling, null checks, data accuracy, and reliable data processing.
- Optimized BigQuery partitioning, clustering, and query execution strategies to improve reporting performance and reduce cloud processing costs.
- Built CI/CD deployment pipelines using Cloud Build, Terraform, Git, and Artifact Registry for automated code deployment and infrastructure provisioning.
- Configured IAM roles, service accounts, and cloud security policies to support secure access management across GCP resources.
- Developed monitoring, logging, and alerting solutions using Google Cloud Monitoring and Cloud Logging for pipeline health checks and operational tracking.
- Collaborated with business analysts, reporting teams, architects, and manufacturing stakeholders to gather data integration and analytics requirements.
- Maintained technical documentation, deployment standards, operational runbooks, and data flow documentation for enterprise data engineering processes.

Environment: Google Cloud Platform (GCP), BigQuery, Cloud Storage (GCS), Dataflow, Dataproc, Pub/Sub, Cloud Composer (Airflow), Cloud Build, Cloud Logging, Cloud Monitoring, PySpark, Python, SQL, Terraform, Git, CI/CD Pipelines, ETL/ELT, Data Modeling, REST APIs, IAM, Service Accounts, Distributed Data Processing, Streaming Pipelines, Batch Processing, Agile/Scrum

Data Engineer

Jun 2019 - Jan 2022

Brio Technologies, Hyderabad, INDIA

Worked as a Junior Data Engineer supporting enterprise cloud migration, ETL development, and analytics initiatives using Google Cloud Platform (GCP) technologies. Assisted in developing and maintaining scalable batch data pipelines and reporting solutions for business analytics, operational reporting, and data integration projects. Supported migration of legacy ETL workflows and on-premise reporting systems into cloud-native GCP data platforms.

Responsibilities

- Assisted in designing, developing, and maintaining ETL/ELT data pipelines using Python, SQL, and PySpark for efficient data integration and processing of structured and semi-structured business datasets.
- Developed SQL queries, transformation logic, and stored procedures for data cleansing, enrichment, and reporting activities.
- Loaded and processed data in BigQuery and Google Cloud Storage (GCS) for downstream analytics and operational reporting applications.
- Assisted in building ingestion workflows for loading data from relational databases, flat files, REST APIs, and cloud storage systems into GCP environments.
- Supported data quality, validation, and reconciliation activities including schema checks, duplicate handling, null validations, data consistency, and accuracy checks using SQL and Python scripts.
- Worked with Cloud Composer (Apache Airflow) workflows for job scheduling, orchestration, and dependency management.

- Monitored scheduled pipelines and assisted in troubleshooting workflow failures, data quality issues, and job execution errors.
- Collaborated with senior engineers, reporting teams, and business analysts to support data integration and analytics requirements.
- Participated in cloud migration activities involving legacy ETL jobs, reporting systems, and database modernization initiatives into GCP platforms.
- Assisted in developing reporting datasets and analytical tables in BigQuery for operational dashboards and business reporting needs.
- Used Git for version control and participated in Agile/Scrum-based development and release processes.
- Supported CI/CD deployment activities using Cloud Build and Git-based deployment workflows across development and QA environments.
- Assisted in preparing technical documentation, deployment procedures, operational runbooks, and support materials for data engineering workflows.
- Participated in production support, incident resolution, and basic Spark and SQL performance tuning activities under senior engineering guidance.

Environment: Google Cloud Platform (GCP), BigQuery, Google Cloud Storage (GCS), Dataflow, Dataproc, Cloud Composer (Airflow), Cloud Build, PySpark, Python, SQL, Git, ETL/ELT, REST APIs, Data Validation, CI/CD Pipelines, Distributed Data Processing, Batch Processing, Data Integration, Agile/Scrum